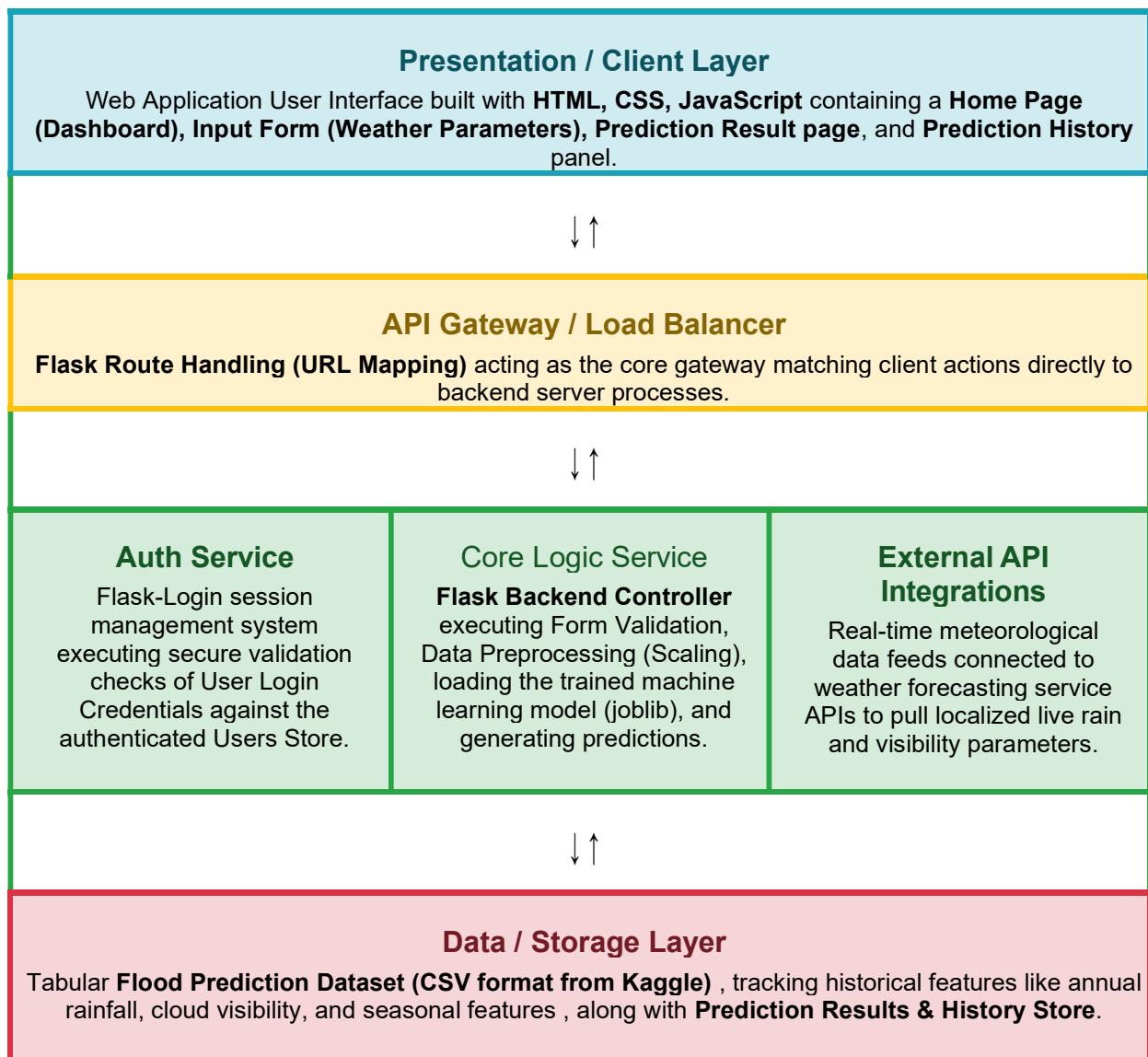


Solution Architecture

Date	28 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	5 Marks

Solution Architecture Diagram:

The Solution Architecture Diagram provides a structured, multi-layer visual roadmap of the RISING WATERS system, mapping the journey of meteorological data across specialized user, presentation, logic, machine learning, data processing, and cloud deployment pipelines. It illustrates exactly how a user web interface connects through a Flask backend to invoke a highly accurate XGBoost prediction engine.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides an intuitive, multi-page frontend dashboard where system users (residents, meteorologists, relief coordinators) can view dashboards, input localized weather parameters, and receive instant alerts.	HTML, CSS, JavaScript, Flask UI templates
API Gateway / Core Logic	Handles incoming traffic routes, maps client URLs, validates input parameter web forms, applies data preprocessing steps, and feeds clean inputs directly into the saved model.	Flask Web Framework, Python
Microservices / ML Layer	Evaluates critical meteorological inputs (annual rainfall, cloud visibility, and seasonal patterns) against the saved classification algorithm to produce a real-time risk status.	XGBoost Classifier, Scikit-learn, Joblib/Pickle, Model (model.pkl), Scaler (scaler.pkl)
Database / Storage Layer	Holds open-source historical climate datasets for preprocessing and baseline metric training, while simultaneously logging transaction requests and historical prediction values.	CSV tabular files, Local Storage System / Cloud Storage instances